LLM-Based Analysis of User Comments for HarmonyOS Applications

AGCommentSpy — From Raw Reviews to Actionable Insights

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Outline

Motivation

HarmonyOS Ecosystem

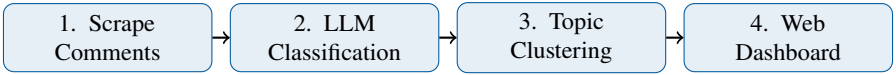
- Huawei's distributed OS — smartphones, tablets, wearables, IoT
- AppGallery hosts **500,000+** native HarmonyOS apps
- User comments are a critical feedback channel for developers

Unique Challenges of HarmonyOS

- **Closed ecosystem:** HDC protocol, limited tooling (no `adb`)
- **Heterogeneous devices:** phones, foldables, watches, in-vehicle
- **Rapid API evolution:** compatibility issues surface in reviews

Existing tools target Google Play / App Store — HarmonyOS needs its own pipeline.

Our Approach at a Glance



Key Insight

Use UI automation to collect reviews + LLM to classify & cluster → transform **unstructured reviews** into **actionable developer insights**.

Contributions

1. **UI-automation crawler** — reliably scrapes AppGallery comments via `hmdriver2` + HDC
2. **LLM classification pipeline** — multi-dimensional sentiment analysis + actionable suggestions
3. **Two-stage clustering** — intra-bucket grouping + cross-batch refinement
4. **Interactive web dashboard** — aggregated results in intuitive visual format
5. **Curated dataset** — **10,000+** comments across 8 categories on HarmonyOS

System Architecture — Overview

Comment Scraping → LLM Classification → Topic Clustering → Web Dashboard

HDC + hmdriver2 | LLM API (JSON mode) | 2-stage LLM cluster | Python HTTP + Vanilla JS

Technology Stack

- `hmdriver2` + HDC — device automation
- `lxml` + XPath — UI parsing
- OpenAI-compatible API — LLM calls
- Python 3.13 — all backend logic
- Vanilla JS + SVG — frontend

Data Flow

- Raw comments → JSON files
- Batch LLM classification (10/batch)
- Classification → clustering
- REST API → dashboard

Stage 1: Comment Scraping

How it works

- Launch AppGallery via deep-link URL
- **Configurable swipes** to load reviews (max 500)
- Parse UI hierarchy (JSON → XML, XPath)
- Extract: username, rating, text, date, device
- **Auto-expand** truncated long comments
- Save per-app JSON files

Scraping Loop (simplified)

```
def scrape_comments(driver, max_swipes=50):
    all_comments = []
    for i in range(max_swipes):
        driver.swipe(...)
        layout = driver.dump_hierarchy()
        xml = json2xml(layout)
        for node in xml.xpath(COMMENT_XPATH):
            c = parse_comment(node)
            all_comments.append(c)
    return all_comments
```

Stage 2: LLM-Based Classification

Goal

Raw app reviews → structured developer insights

Classification

- **Sentiment:** Positive, Negative, Neutral
- **7 dimensions:**
  - Functional Experience
  - System Compatibility
  - Stability & Performance
  - UI & Interaction
  - Operations & Policies
  - Appreciation
  - Content Ecosystem

Structured Output

- Pain point + actionable suggestion
- Comment ID for traceability

Reliability

- JSON schema, 10 comments/batch
- Auto-repair malformed responses

Example

Raw review

"The HarmonyOS version crashes when opening videos, and landscape mode is poorly adapted."

LLM classification output

- **Sentiment:** Negative
- **Dimension:** Stability & Performance
- **Pain point:** Video playback crashes
- **Suggestion:** Check crash logs; improve landscape adaptation

Stage 3: Two-Stage Topic Clustering

Initial attempt: word-embedding clustering

- Sentence-BERT + KMeans/DBSCAN
- **Failed:** same root cause, different wording → far apart in vector space → split into different clusters

Our approach: LLM-as-clusterer

- **Stage 1:** partition by (dimension, sentiment); LLM clusters items within each bucket in batches of 18
- **Stage 2:** merges overlapping clusters across batches and standardizes topic names

Example

cluster_name: 好友信息功能缺失
canonical_issue: 鸿蒙版微信缺少好友来源、共同群聊、添加好友时间等常用功能
size: 1

#264 华为畅享*90 Pro Max*
"好友来源、好友共同群聊、添加好友时间等都没有，求更新"

#272 *HUAWEI Pura 70 Pro+*
"连朋友资料都没，版本比别人低..."

Stage 4: Web Dashboard

Four Tabs

1. **Overview**
   - Total comments, classified count
   - Donut chart (sentiment)
   - Bar chart (dimensions)
2. **Comments**
   - Paginated list + full-text search
   - Username, rating, date, device
3. **Analysis**
   - Dimension-filtered pain points
   - LLM-generated **actionable suggestions**
4. **Clusters**
   - Topics by dimension $\times$ sentiment
   - Expandable sample members

Tech Stack

Vanilla JS (no framework) + inline SVG charts
Python HTTP server + REST API endpoints

Results — Dashboard Overview

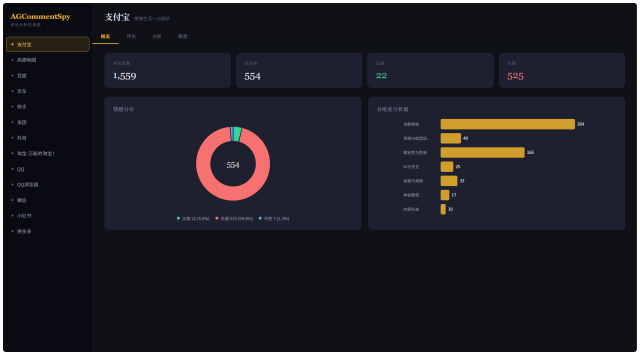


Figure: Overview — aggregate statistics and charts

- **Dataset: 10,000+** comments across **45+** apps in 8 categories
- **Sentiment distribution** visualized as an SVG donut chart
- **Dimension breakdown** shown as a horizontal bar chart
- One-click access to total/classified counts

Results — Analysis & Clusters

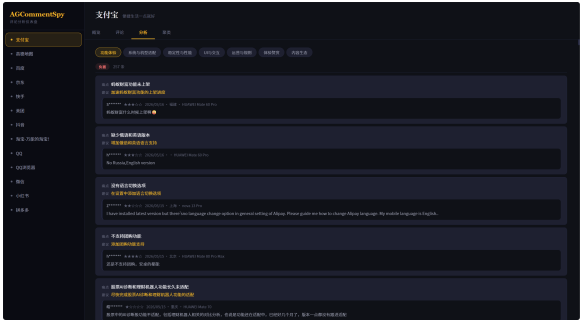


Figure: Analysis — per-dimension issues with suggestions

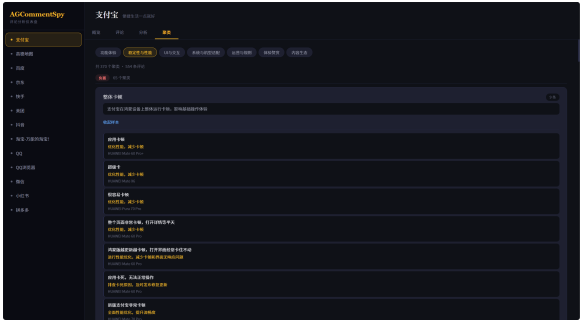


Figure: Clusters — grouped issues with canonical descriptions

Evaluation & Discussion

Classification

- 7 dimensions balance **granularity** with **actionability**
- Every negative issue → concrete `Actionable_Suggestion`

Clustering

- Word-embedding clustering separated same-root-cause items (lexical, not semantic)
- LLM-as-clusterer reasons about root causes, merging despite vocabulary variance
- Two-stage design: focused sub-problems + lightweight reconciliation

Conclusion

What We Built

AGCommentSpy — end-to-end pipeline that transforms **unstructured AppGallery reviews** into **actionable developer insights** via UI automation + LLM-based NLP.

Key Takeaways

- UI automation compensates for the **lack of public APIs** on HarmonyOS
- LLMs enable **zero-shot** fine-grained classification of user reviews
- Two-stage clustering turns **scattered comments** into **coherent topics**
- Lightweight web dashboard makes results **accessible to developers**

Future Work

1. **Automated labeling** — human-annotated gold standard for evaluation
2. **LLM comparison** — benchmark multiple providers on cost-quality
3. **Incremental scraping** — only scrape new comments since last run
4. **Developer workflow integration** — auto-file findings to GitHub Issues / Jira

Thank You!

Questions & Discussion

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